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Heat Pump Feasibility Study

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Dear Professor Collins,

This report, entitled Heat Pump Feasibility Study, was written to fulfil my 3A work term report requirements. This is my 2nd work report submission.

The basis of the project was in response to a discussion between myself and my father. As a result of high inflation, utility costs had skyrocketed this winter season which meant heating costs would eat into the family businesses profit margin. Because of this, he had expressed his potential interest in alternative heating solutions.

Currently, Richters Herbs operates several propane unit heaters to keep the greenhouse facilities at a relatively stable temperature throughout the winter months. In order to relieve some of the financial pressure, I took it upon myself to conduct a study to determine the feasibility of installing and operating alternative heating methods.

This report was written entirely by me, and has not received any previous academic credit at this or any other institution. I would like to acknowledge the help of Conrad Richter, President of Richters Herbs, who defined the purpose of the project and provided all of the information I would need to conduct my analysis. My role in the project was to collect and analyze the data I would need, conduct further research, and write the report. The project lasted 3 months. Conrad Richter can be contacted at (905) 640-6677.

Best Regards

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Executive Summary

As the world continues to change culturally, politically and even environmentally, one aspect of the Canadian experience which remains constant for the time being are the long and cold winters. It is for this reason all Canadian households and businesses are required to heat their property for 4-5 months every year. To do this, Canadians rely on various methods to battle the cold. In Ontario, most use electricity to heat their spaces, while others use fossil fuels, with a small minority using renewables [1].

Along with rising inflation and increasing carbon taxes, these fossil fuels such as propane will continue to rise in price over the coming years. For this reason, certain individuals and businesses, such as Richters Herbs, are choosing to adopt new methods to heat their spaces. One of the most popular methods in recent years is the transition to electrical heat pumps. Electrical heat pumps generally offer good heating performance and good efficiency at average winter temperatures and along with the relatively low cost of electricity when compared to fossil fuels, heat pumps generally reduce operational costs [5].

Following this logic Richters Herbs, the client, aims to reduce the overall operational costs of its heating system in its greenhouse facilities to increase profit margins while minimizing the required capital investment. To do this the client has requested a feasibility analysis to determine the optimal implementation of an alternative heating source which satisfies these requirements.

After carefully analyzing possible solutions, this report has concluded that the client will need to install at least 24 Lennox ELP/ELS120 electrical heat pumps at a total capital cost of \$240 000 to act as the primary source of heating with the legacy propane unit heaters acting as a backup, only to be activated in the most extreme conditions. With these new heat pumps the client can be expected to save around \$45 989 in operational costs annually, which results in a period of approximately 5.5 years to break-even.

By way of further analysis, this study aims to decrease operational and investment costs beyond these initial estimates through an additional Computational Fluid Dynamics (CFD) simulation, along with the application of government tax incentives and possible loans and grants.

1.0 Introduction

One of the unavoidable realities of residing in Canada are the cold and dark winter months. While climate change presents the possibility of altering this, for the foreseeable future Canadian winters will continue to provide an annual hurdle to all its citizens. A large portion of this hurdle comes from the combination of both residential and commercial heating requirements. Not only is this heating logistically taxing, it also has an extraordinary financial cost. In 2015, Ontarians consumed 522 205 592 Gigajoules (GJ) of energy for heating or 100.2 GJ per household [1]. This equates to \$2317 per household on average, or approximately 3% of disposable income [2].

1.1 Commercial Heating

The commercial sector also sees high costs associated with heating. The focus of this report, Richters Herbs, spends approximately \$110 000 annually on heating currently. Richters Herbs is a small business based in southern Ontario which grows and sells a vast catalog of herbs both domestically and internationally (Figure 1). Richters Herbs operates all year and therefore must keep its facilities at a relatively constant minimum ambient temperature to keep the plants healthy.



Figure 1 - Greenhouse number 1 [3]

To do this, Richters Herbs currently uses propane-fired unit heaters placed in strategic locations throughout its facilities. Due to rising inflationary costs, as well as increasing carbon taxes implemented by the Canadian government, the \$110 000 spent on propane annually is expected to continue to rise with each new winter season [4].

1.4 Objective

As a result, it is the objective of this report to analyze the current heating system employed by the client and perform a feasibility analysis on alternative heating methods.

2.0 Problem Definition

At Richters Herbs, heating is required to keep the minimum ambient temperatures of specific zones at specific temperature values throughout the year. There are three distinct temperature zones in the facility. Greenhouse number 3 serves as a propagation area which promotes the growth of young cuttings. As a result, this zone requires a slightly higher minimum ambient temperature of 60° F. The boiler room serves as a boiler room, a packing/loading facility, as well as tool storage. Because no plants reside here, the minimum ambient temperature is not of concern and the room is heated by the adjacent greenhouses. The final and largest zone encompasses all of the remaining greenhouses combined, which require a minimum ambient temperature of 50° F.

Currently, there is a total of 14 Modine PDP200 propane-fired unit heaters located throughout the greenhouse facilities which output $166\,000 \frac{BTU}{hr}$ each (Figure 2). There are two heaters in greenhouse number 3 and the remaining 12 heaters are spread out in the other structures, excluding the boiler room. Any time various temperature sensors throughout the facility detect an ambient temperature below their respective target temperature, the adjacent heaters will activate until the target temperature in that area is reached.

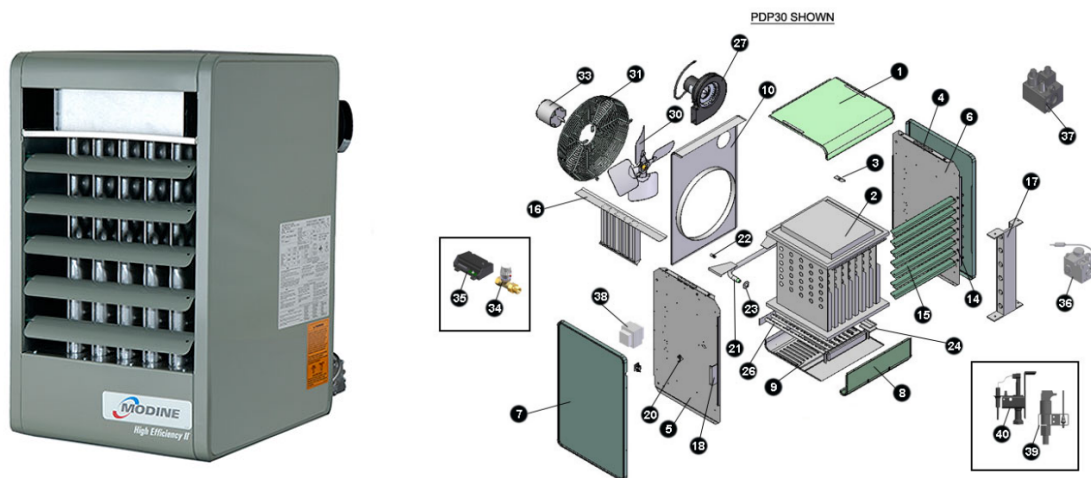


Figure 2 - Modine PDP200 Propane Unit Heater [5]

The driving factor in the operating costs of these heaters is the cost of the propane. Each heater consumes approximately \$66 of propane daily which adds to around \$924 of propane consumed by the whole system every day. These numbers are based off historical data (2018, 2019, 2020) however, they do not account for the ever-rising costs of propane due to inflation and added carbon taxes. For example, Richters expects a huge increase in propane prices of around 100% for the 2021-22 winter season.

2.1 Constraints and Criteria

With such large increases in utility costs anticipated for this season and for the seasons to come, an alternative heating method which reduces operational costs and increases the overall profit margin while maintaining the minimum required ambient temperatures in all but the most extreme cases is desired. The capital investment associated with the installation costs should also be minimized.

3.0 Results and Analysis

3.1 Survey

The first step in performing an analysis of the heating system of the greenhouse facilities at Richters Herbs was to perform a comprehensive survey. With the help of the greenhouse manager, an interior and exterior walkthrough of all the facilities was conducted. This survey aimed to gather data on a few critical metrics which would be useful later in calculating the heating load requirements. The first was the physical dimensions of the greenhouses. Dimensions were gathered in feet with a measuring tape, with an error tolerance of $\pm 1/8$ feet.

The next important item to catalog was the physical materials and overall construction of the various facilities. The majority of the facilities utilize a simple frame constructed of hollow galvanized steel tubes covered in an air-filled double layer of polyethylene (Figure 3). The interior of these polyethylene greenhouses is mostly empty besides the support beams and the benches containing the plants, while the exterior shape is generally a rectangular prism with circular roofs. Greenhouses numbered 8 and 3 are notable exceptions, with circular walls and roofs.



Figure 3 - Greenhouse Number 4 Exterior



Figure 4 - Greenhouse Number 8 Interior



Figure 5 - Greenhouse Number 3 Interior

Utilizing a galvanized steel frame with tempered glass panes, greenhouse number one contains several benches and forms a rectangular prism with a steep triangular roof construction (Figure 6). Finally, the boiler room has a painted steel exterior shell which is backed by four inches of interior insulation (Figure

7). The boiler room contains a boiler, generator, a few tables, fridges, and tool cabinets. Similarly to greenhouse number 1, the exterior of the boiler room forms a rectangular prism with a triangular roof.



Figure 6 - Greenhouse Number 1 Interior



Figure 7 - Boiler Room Exterior

The larger polyethylene greenhouses (excluding numbers 8 and 3) also contain large exterior bay doors in combination with smaller exterior swinging doors (Figure 8). The bay doors vary from 9 ½' x 12' to 7 ½' x 12', while the swinging doors are generally 8' x 4'. These doors all consist of insulated painted steel frames with acrylic windows. The boiler room contains a 10' x 12' painted steel, segmented garage door with acrylic windows. The north-end greenhouse also has a non-insulated vestibule entrance.



Figure 8 - Example of Bay Door on Greenhouse Number 4

The last item included in the survey were the heating elements along with pictures of their respective nameplates (Figure 9). The locations of the heaters were noted. 14 heaters were counted in total.

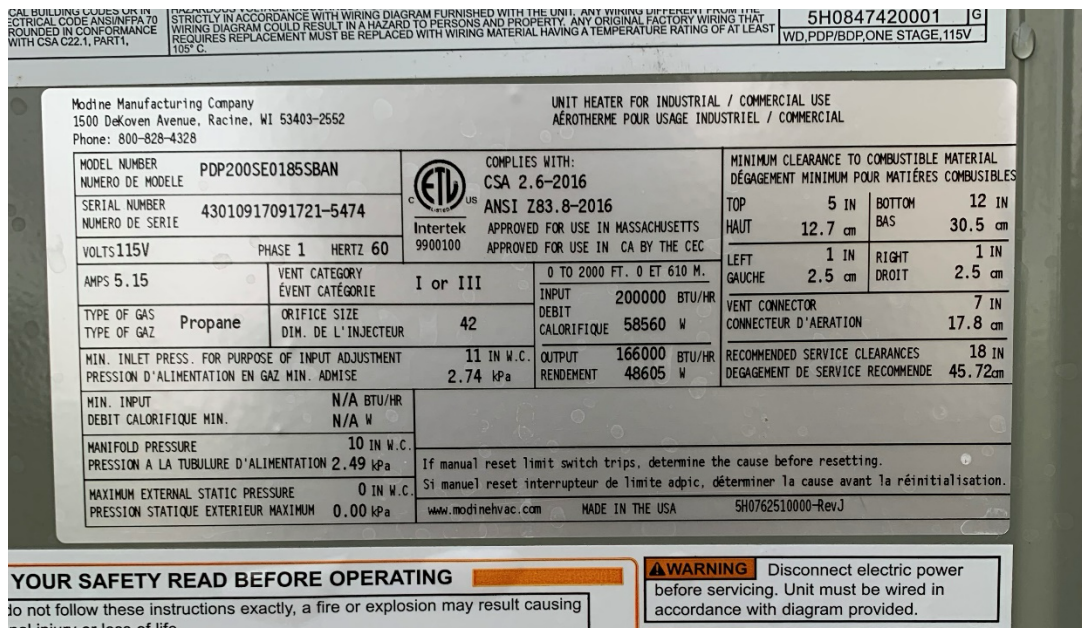


Figure 9 - Example of Nameplate on Unit Heater in Greenhouse Number 3

3.2 Modelling

After completing the survey of the greenhouse facilities, the next step was to construct CAD models of the structures. The first model was a 2D floorplan of the facilities in AutoCAD which included the locations, size and types of doors; the locations and technical specifications of the unit heaters; the names/numbers of the different facilities; and interior walls (Figure 10). This floorplan would act as a resource for determining the locations and dimensions of any item during later calculations as well as a starting reference for the following model.

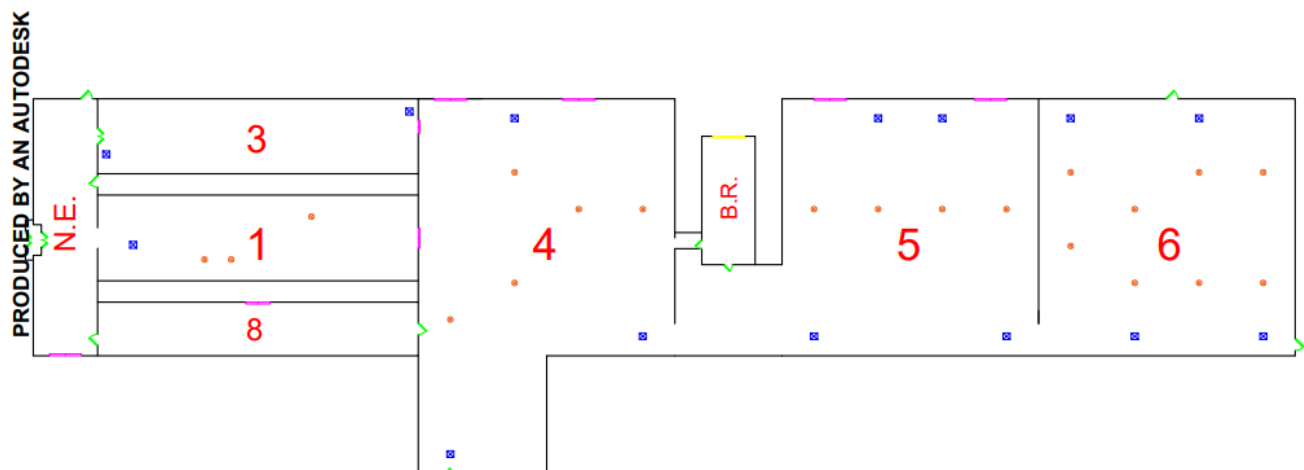


Figure 10 - Floorplan of Facilities at Richters Herbs

Using the floorplan and the remaining data collected during the survey, a 1:1 3D CAD model of the facilities was created using Fusion 360 (Figure 11). The model included all exterior items such as doors and physical materials; it would act as a resource to determine the individual and overall surface areas and volumes of the facilities which is an integral part of the mathematical models used later.

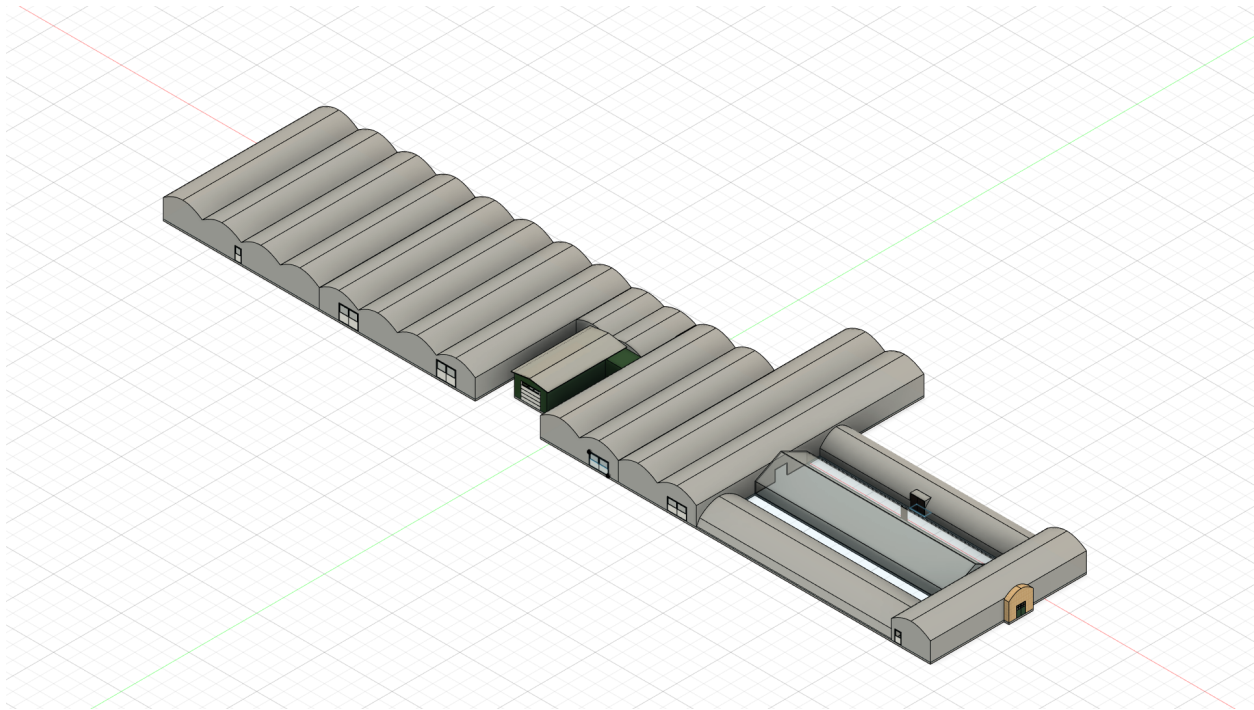


Figure 11 - Isometric View of North-East Side

3.3 Alternative Heating Method

To reduce operational costs while increasing the overall profit margin, an alternative to propane unit-heaters is required. There are many options which abide by these criteria. There are other fossil fuel based heaters such as oil and natural gas heaters which would reduce costs, albeit by a relatively small margin [6]. There are also renewable sources such as geothermal heat pumps which reduce operational costs by a significant margin but require extensive restructuring of the current heating system along with significant infrastructure additions [6]. To abide by the listed criteria as well as the constraint of keeping capital investment costs at a minimum, the most attractive option is the electrical heat pump. These heat pumps provide good performance at most expected winter temperatures with relatively small electrical power consumption. They are also relatively simple to install and don't require any major infrastructure modifications. Additionally, they have the added benefit of providing a more sustainable and less environmentally impactful effect considering the sources of electricity in southern Ontario [7].

With this in mind, the proposed heat pumps must be both effective and efficient in colder temperatures; they must be relatively small, fully contained units; and they must be relatively cheap and simple to install. Following these requirements, the Lennox ELP/ELS120 Elite® Series of Large Split Systems are an acceptable choice (Figure 12) [8].



Figure 12 - Lennox ELP/ELS120 Heat Pump [8]

3.4 Calculations

After collecting the necessary data, the next step was to compute the mathematical models in order to quantitatively understand the heating load requirements, the current heating system, and to propose an alternative heating solution. The basis for the models used is “NONRESIDENTIAL COOLING AND HEATING LOAD CALCULATIONS” in the ASHRAE Fundamentals Handbook which is compliant with the Air Conditioning Contractors of America (ACCA) guidelines.

3.4.1 Heating Load

The first calculation to consider are the heating loads for the various structures at Richters. There are three variables in a basic heating load model. There is a heat transfer coefficient which is unique to every material and is dependent on the configuration of said material. The heat transfer coefficient is measured in units of $\frac{W}{m^2K}$ or $\frac{BTU}{ft^2°Fhr}$. The other variables are exterior surface area in m^2 or ft^2 and the temperature differential between the exterior and interior temperatures in units of Kelvin or Fahrenheit. The resulting calculated value represents the heating load of the respective structure in units of W or $\frac{BTU}{hr}$ (Equation 1).

Equation 1 - Example of Heat Load Calculation of Greenhouse Number 1 at 15 Degrees F

$$Q = U \cdot A \cdot \Delta T$$

$$Q_1 = \left(0.607 \frac{BTU}{ft^2 \cdot ^\circ F \cdot hr}\right) (54434 ft^2) (50 - 15^\circ F)$$

$$Q_1 = 124343 \frac{BTU}{hr}$$

While the surface area and heat transfer coefficients remain constant for any given structure amongst the facilities, the temperature differential changes depending on the conditions of the model. For the sake of simplicity, heating loads are calculated at three different exterior temperatures. These exterior temperatures reflect the average daily temperatures of three different time periods in southern Ontario [9]. The first is 15° F which represents the average daily temperature for the months of January and February. The next is 23° F which represents the average daily temperature for the months of December and March. Finally, 32° F represents the average daily temperature of approximately a third of November and April. The interior temperatures are to be kept at a target minimal ambient temperature of 60° F in greenhouse number 3 and 50° F in the other structures in accordance with the given requirements (Table 1).

Table 1 - Calculated Heating Loads at Various Temperatures

Exterior Temperature (°F)	Heating Load ($\frac{BTU}{hr}$)	Heating Zone
15	1333633	Remaining Greenhouses
	124342.7	Greenhouse #1
	135746.1	Greenhouse #3
23	1028803	Remaining Greenhouses
	95921.54	Greenhouse #1
	111613.5	Greenhouse #3
32	685868.4	Remaining Greenhouses
	63947.69	Greenhouse #1
	84464.24	Greenhouse #3

From the calculated heating loads it is noted that across all exterior temperatures, the majority of the heating load comes from the polyethylene greenhouses as they account for the majority of the facilities

overall surface area. The next is the polyethylene greenhouse number 3, followed closely by the glass greenhouse number 1 as a result of its lower heat transfer coefficient which contains heat with greater efficiency (Figure 13).

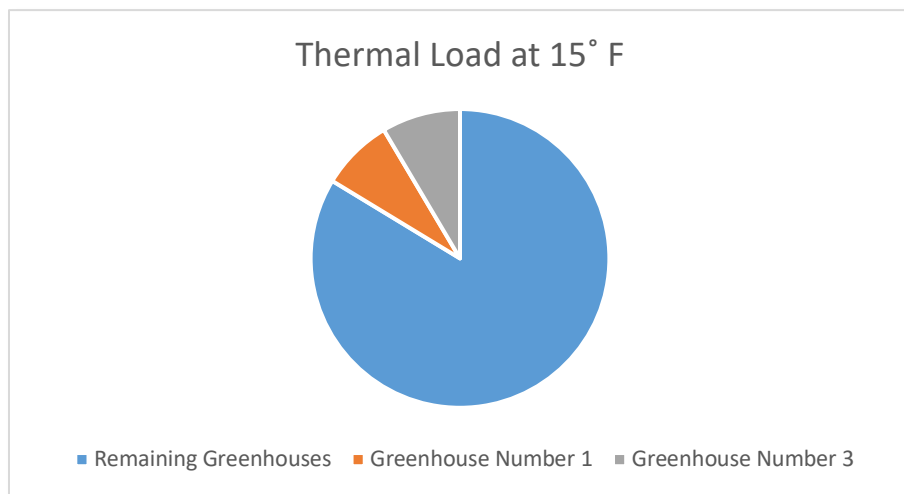


Figure 13 - Calculated Thermal Loads of Various Zones at 15 degrees F

3.4.2 Legacy Heating Costs

In order to determine possible alternative heating solutions, it is necessary to first compute the current energy usage/cost. Based off the analysis of the financial statements of the 2020 fiscal year at Richters, it is estimated that approximately 71.877% of utility costs result from propane usage. Multiplying this factor by the overall utility costs in 2020 results in an approximate total propane cost. Applying this same percentage to previous years (2018 and 2019), and calculating the mean, produces a good estimate for the historical average annual propane cost. The outcome of this calculation is a yearly total average of \$110 909, which equates to around \$924 daily when considering a heating season of 4 months.

3.4.3 Heat Pump Energy Consumption

Before calculating the operating costs of the proposed heat pumps, the electrical power draw must be determined. The first step in determining the power draw is to determine the amount of heat pumps that will be required. Based off the heating loads calculated earlier, it is assumed that the lowest exterior temperature is 15° F. At this exterior temperature, the ELP/ELS120 series heat pumps output $67\,900 \frac{BTU}{hr}$ (Table 2). Dividing the respective heat loads of the three systems (greenhouse numbers 1, 3 and the rest) by the heat pump output results in the required number of heat pumps for that system.

Table 2 - Performance of Lennox ELP/ELS120 Heat Pump at Various Exterior Temperatures [8]

ELP120S4S + ELA120S4D		
HEATING PERFORMANCE at 4000 cfm Indoor Coil Air Volume		
Temperature °F	Motor Input kW	Output kBtuh
65	7.56	141.1
60	7.4	133.8
55	7.24	126.5
50	7.08	119.1
47	6.98	114.8
45	6.92	111.8
40	6.75	104.5
35	6.59	97.2
30	6.43	89.9
25	6.27	82.6
20	6.11	75.2
17	6.02	70.8
15	5.96	67.9
10	5.8	60.5
5	5.43	54
0	5.07	47.4
-5	4.7	40.9
-10	4.34	34.3
-15	3.98	27.8
-20	3.61	21.2

The required input power (*kW*) can then be multiplied by the number of heat pumps in the respective system which results in the total power draw at 15° F. This process is then repeated at the different exterior temperatures (23 and 32° F) for all the systems. The resulting data encompasses the power draw of all the proposed heat pumps at their respective operating conditions in *kW* (Table 3).

Table 3 - Hourly Power Draw of Heat Pump System

Exterior Temperature (°F)	Heating Zone	Number of Heat Pumps	Power Draw (kW)
15	Remaining Greenhouses	20	119
	Greenhouse #1	2	11.9
	Greenhouse #3	2	11.9
23	Remaining Greenhouses	20	105
	Greenhouse #1	2	10.1
	Greenhouse #3	2	10.9
32	Remaining Greenhouses	20	94.0
	Greenhouse #1	2	8.68
	Greenhouse #3	2	9.87

For example, for the glass greenhouse number 1, start by dividing the calculated heat load at 15° F by the tabulated performance of the ELP/ELS120 at the same temperature. The resulting numerical value should be rounded up to determine the minimum number of heaters (2) required to heat this system to its minimum ambient temperature at the lowest exterior temperature. Using the same performance data, multiply the power draw of the heat pump at this exterior temperature by the number of heaters required which results in the overall hourly power draw of the system. For the next exterior temperature, divide the heat load by the number of heaters to get the required heat output for each heater at 23° F. Interpolate the power draw of the heat pump at this value and multiply by the number of heaters to determine total power draw. Repeat this process for the exterior temperature of 32° F and do the same for greenhouse number 3 and the remaining facilities.

3.4.4 Heat Pump Costs

The final step in calculating the total cost of the proposed ELP/ELS120 heat pump system is to estimate the total annual power draw of the system to then calculate the electrical cost. As stated previously, the average daily temperatures of January and February in southern Ontario are 15° F. For this reason, first add the hourly power draw of the all the systems at 15° F and multiply that value by the total hours in the January and February (31 + 28 days = 59 days × 24 hours = 1416 hours total). The resulting value represents the total power draw in *kWh* of the heat pumps during January and February (Equation 2). Repeat this process for the months of December and March, which have a daily average temperature of 23° F, and a fraction of both April and November (20 days), which have a daily average temperature of 32° F.

Equation 2 - Total Power Draw of Heat Pumps During January and February

$$W_{el} = (119 + 11.9 + 11.9)kW \times (1416 h)$$

$$W_{el} = 202205 kWh$$

Finally, sum the calculated power draws to find the total annual power draw (444 000 *kWh*) of the entire heat pump system. With this value it is possible to compute the estimated electrical cost of the system by

multiplying the total power draw by the unit cost of \$0.115 per *kWh* in southern Ontario [10]. The resulting value of \$51 060 represents the annual cost of operating the 24 heat pumps in the proposed alternative heating system.

It should be noted that while this value represents the operational cost of the heat pumps in the alternative heating system, this is not the only operational cost observed. By definition, the efficiency of heat pumps decreases as the exterior temperature decreases [11]. At temperatures below 15° F, the efficiency of the ELP/ELS120 heat pumps decreases significantly. To avoid acquiring additional units to compensate for these fringe cases which would take up additional space and would add to the total acquisition cost, the legacy propane unit heaters will activate to supplement the heating capacity of the heat pumps under these circumstances. Considering an estimate of around 15 days a year in which the average daily temperature falls significantly below this value ($\sim -5^{\circ}$ F), the additional cost of operating the legacy propane heaters can be determined. Assuming the daily cost of this operation is similar to the previously calculated current daily cost of operation of \$924, the total additional cost is 15 days $\times$ \$924 which results in \$13 860. Adding this value to the cost of operating the heat pumps, the total operating cost of the proposed heating system comes to \$64 920.

3.5 Cost Comparison

After estimating both the current heating costs associated with the propane heating system and the proposed costs of the combined heat pump and propane system, the dollar values can be compared to determine the feasibility of the transition. By subtracting the average annual cost of the legacy propane system (\$110 909) with the annual cost of the novel system (\$64 920), it is revealed that the heat pump system would save approximately \$45 989 in operational costs every year.

To determine the time required to break even on such a large-scale capital investment, the unit cost of acquiring and installing the 24 heat pumps is necessary. Assuming the cost for each heat pump is around \$3500 and the installation cost is around \$6500 each, which is likely much higher than reality, the total cost of all heat pumps is \$240 000. Dividing this number by the yearly saved operational cost results in an estimated time to break-even on the investment of around 5.5 years.

4.0 Conclusions

4.1 Conclusions

In conclusion, the proposal to acquire 24 Lennox ELP/ELS120 heat pumps to act as the primary heating source for Richters Herbs facilities with the legacy propane heaters acting as a backup heat source in all but the most extreme cases, meets the initial constraints and criteria of the project. The electrical heat pumps reduce operational costs by approximately half which in turn has the effect of increasing profit margins while also keeping installation costs at a minimum, with a break-even time of around 5.5 years. The heat pumps also have the unintended effect of reducing the carbon footprint of Richters Herbs operation which will only serve to benefit all stakeholders involved by decreasing environmental damage and improving overall public reception.

4.2 Next Steps / Recommendations

Looking forward, there is still more work to be done to improve the effectiveness and efficiency of the new heating system to further reduce operational and investment costs to improve profit margins. While the simple mathematical models used to estimate the use of the heat pumps is a good starting point in providing the client with an idea of the scale and effect of the transition, further analysis is required to provide a thorough and more accurate proposal. Using the 3D CAD model created in combination with the surveyed data collected, a comprehensive Computational Fluid Dynamics (CFD) simulation should be conducted. This simulation will provide the client with the best possible solution to their heating requirements including the proper number and placement of the heat pumps to ensure the maximized efficiency of the overall system. The CFD simulation could also provide the client with other unforeseen solutions which could have a similar effect as the heat pumps in other ways such as modifications to the interior and exterior structures.

Along with the CFD simulation, another additional form of analysis which could serve to further decrease the capital investment of installing the heat pumps which would decrease the time to break-even, providing an earlier increase in profit margins, is the possibility of government assistance programs. For example, the federal government currently offers tax incentives to businesses which aim to transition to renewable/clean energy consumption [11]. Conducting additional research into the conditions and effects of such tax incentives could serve to further reduce initial investment costs and even operational costs. There is also the possibility of additional grants and loans that could be available through similar programs which should be included in this additional research.

With these two areas of further investigation, there remains much work to be done. With these strategies in mind, it is clear that even greater results are still possible which will serve to further satisfy the clients initial requirements.

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